THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 12: **Reproduction: How Life Continues** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Plants raised by vegetative propagation are genetically identical to the parent. Reason (R): They are produced by mitosis from a single parent. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A woman's menstrual cycle is 28 days and menstruation begins on 1 March. The likely day of ovulation is: 1

(a) 1 March **(b)** 28 March **(c)** 14 March **(d)** 7 March

3. A student places a potato piece with an eye in moist soil and another without an eye. The result will be: 1

(a) the one without an eye grows **(b)** the one with an eye grows
(c) both grow **(d)** neither grows

4. Seeds dispersed by wind have: 1

(a) no coat **(b)** fleshy covering **(c)** wings or hairs **(d)** hooks

5. The female reproductive structure that is the site of fertilisation and passage of the egg is the: 1

(a) uterus **(b)** oviduct **(c)** cervix **(d)** vagina

6. Assertion (A): Cross-pollination leads to variation. Reason (R): It involves the fusion of gametes from different plants. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. The egg released on 14 March remains fertilisable for about 1 to 2 days. The most fertile days are: 1

(a) 28 to 30 March **(b)** 1 to 5 March **(c)** 13 to 15 March **(d)** 20 to 25 March

8. The grafting is successful when the: 1

(a) leaves are cut **(b)** xylems are far apart
(c) cambia of stock and scion are in contact **(d)** roots are removed

9. **Assertion (A):** Offspring produced by asexual reproduction are genetically identical to the parent. 1
Reason (R): Asexual reproduction involves a single parent and no fusion of gametes.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Yeast reproduces by budding. 1
Reason (R): *Amoeba* reproduces by binary fission.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A cell in the human body has 46 chromosomes. State the number of chromosomes in a muscle cell, sperm, egg and zygote. 2

12. Explain the role of the pollen grains and the ovule in seed formation. 2

13. State two modern methods of assisted reproduction and their purpose. 2

14. Why does a potato sprout in a dark cupboard? Which structure is responsible? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A woman has a 30-day menstrual cycle with menstruation starting on day 1. Calculate the probable day of ovulation, and the fertile period (taking the egg life as 1 day and sperm life as 3 days). 3

16. A young plant grown from a seed shows features different from the parent while a plant grown from a stem cutting does not. Explain. 3

17. Explain the different methods adopted by plants to prevent self-pollination. 3

OR

Compare the reproductive strategies of a frog, a hen and a human as to number of eggs, parental care and survival of young. 3

18. Why do fruit growers prefer grafting to growing from seeds? Give three reasons. 3

OR

State three effects of a high birth rate on the family and on the nation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rakesh observes a drop of pond water under a microscope. He sees an *Amoeba* whose nucleus has divided and the cell is pinching into two daughter cells. In another slide, he sees yeast cells with small outgrowths, and on a damp bread slice at home he finds a fuzzy growth of mould. 5
(i) Name the method of reproduction seen in *Amoeba*.
(ii) Name the method shown by yeast.
(iii) How do bread moulds multiply?
(iv) A single *Amoeba* divides every hour. How many *Amoeba* are produced from one after 4 hours?

20. Explain the male and female reproductive systems, gamete formation, fertilisation, implantation, development of the embryo, role of the placenta, and birth with the names of the structures. 5

OR

Explain how the reproductive health is affected by lifestyle, nutrition, sexually transmitted infections, and substance abuse and the measures to maintain it. 5

END OF QUESTION PAPER — Total: 40 Marks